

PATTERN RECOGNITION AND MACHINE LEARNING

HYEONWOO YU

Dept. of Intelligent Robotics, SKKU

2025 2nd Semester

Notice

Lecture Notes: Slides

References:

- Machine Learning: A Bayesian and Optimization Perspective, Sergios Theodoridis, Academic Press, 2020
- Pattern Recognition and Machine Learning (Information Science and Statistics), Christopher Bishop, Springer, 2006

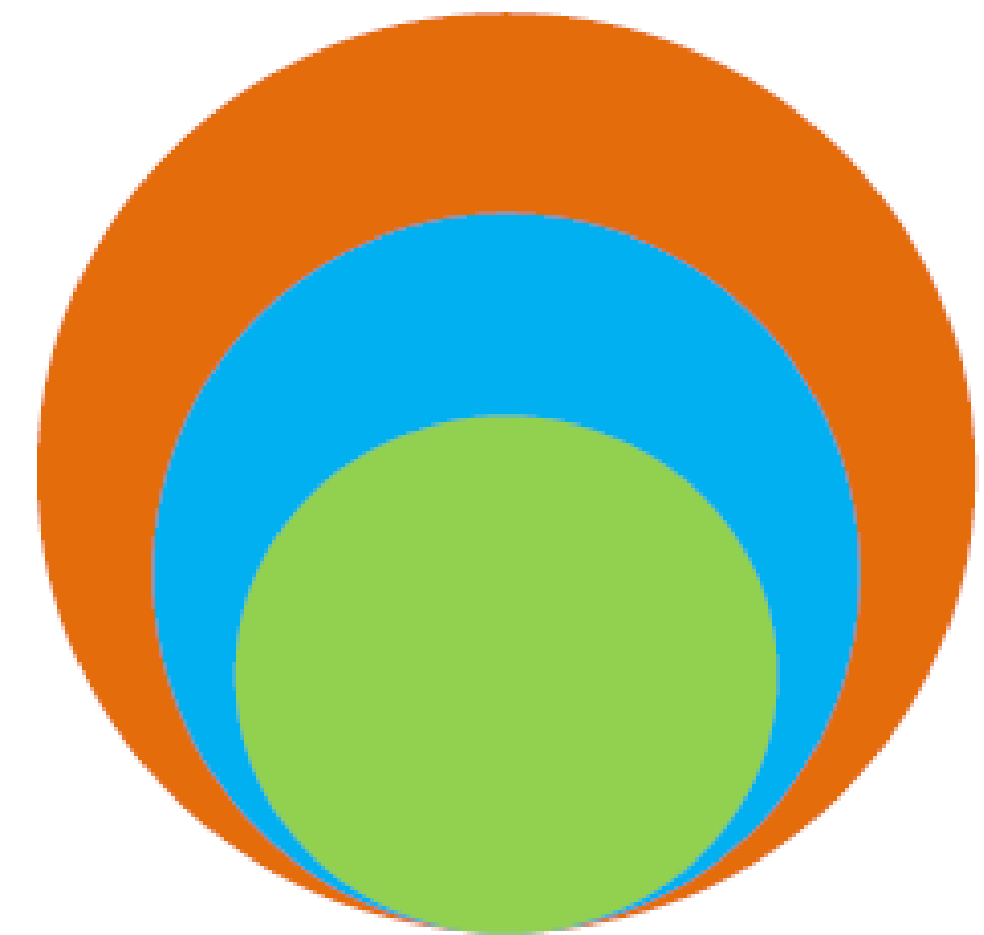
Evaluation: Attendance 10%, Homework 20%, Midterm 30%, Final 40%

Class web: icampus

Office: 23338

INTRODUCTION TO

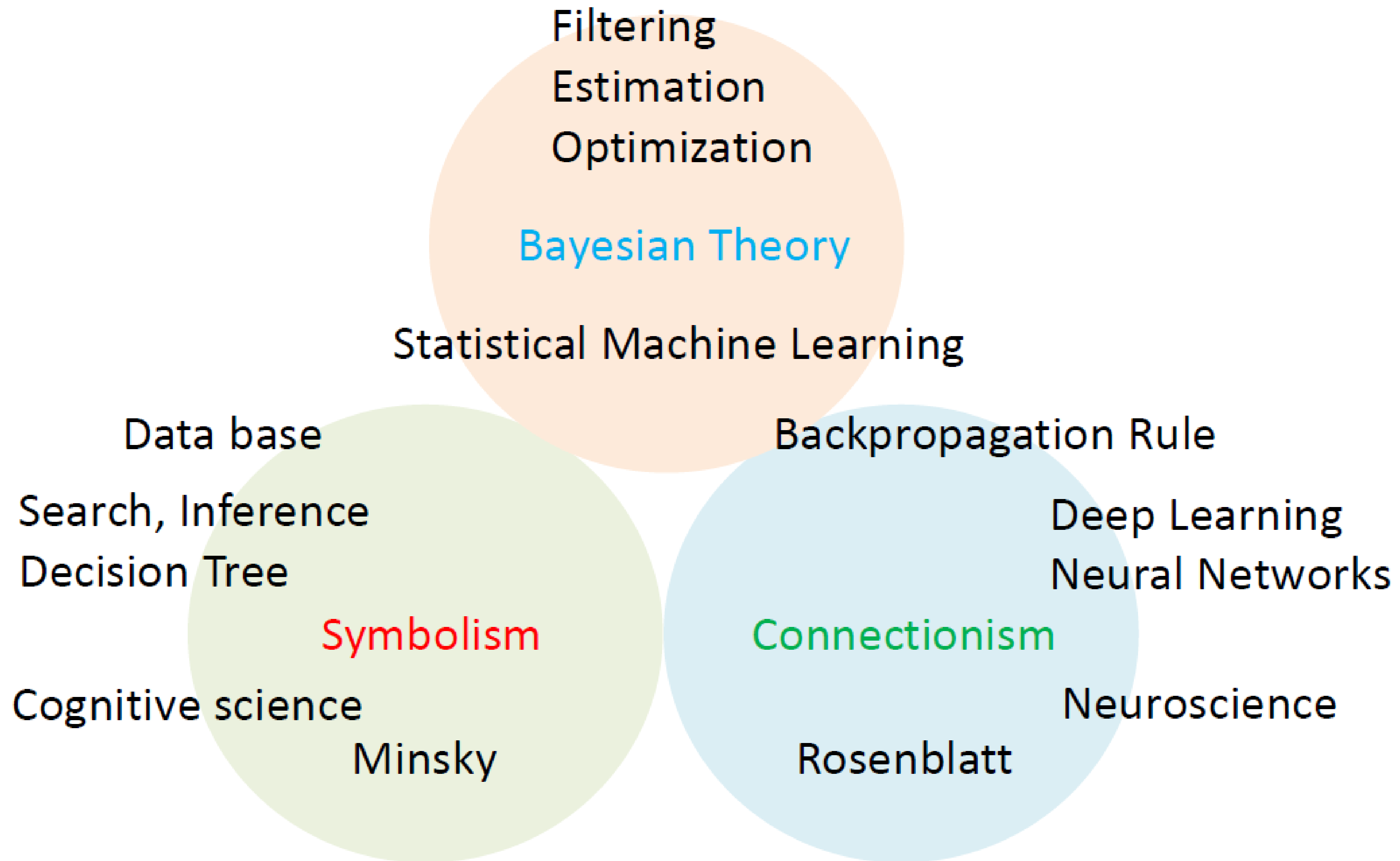
AI: ARTIFICIAL INTELLIGENCE
ML: MACHINE LEARNING
DL: DEEP LEARNING



HYEONWOO YU

Dept. of Intelligent Robotics, SKKU

Artificial Intelligence



Artificial Intelligence

Learning from Experience (Observations, Examples)

?

Inference(Reasoning) for a Question

?

Artificial Intelligence

Learning from Experience (Observations, Examples)

If birds are given, then we can learn their features such as # of legs, shape of mouth, etc.

If cancer patients are given, then we can observe their symptoms via diagnosis

.....

Inference(Reasoning) for a Question

If features of something are given, then we can recognize what is it.

If symptoms of a patient are given, then we can infer what is his decease.

.....

Artificial Intelligence

Learning from Experience (Observations, Examples)

If birds are given, then we can learn their features such as # of legs, shape of mouth, etc.

If cancer patients are given, then we can record their symptoms via diagnosis

If $y = y_1$, then $x = x_1$

If $y = y_2$, then $x = x_2$

y			x			
y_1	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	x_{16}
y_2	x_{21}	x_{22}	x_{23}	x_{24}	x_{25}	x_{26}

DB
Decision Tree

Inference(Reasoning) for a Question

If features of something are given, then we can recognize what is it.

If symptoms of a patient are given, then we can infer what is his decease.

If $x = x_1$, then $y = y_1$

If $x = x_2$, then $y = y_2$

y			x			
y_1	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	x_{16}
y_2	x_{21}	x_{22}	x_{23}	x_{24}	x_{25}	x_{26}

Symbolism

Search-based Inference Engine

Artificial Intelligence

Learning from Experience (Observations, Examples)

If $y = y_1$, then $x = x_1$

If $y = y_2$, then $x = x_2$

$$p(x = x_i | y = y_j), p(y = y_j)$$

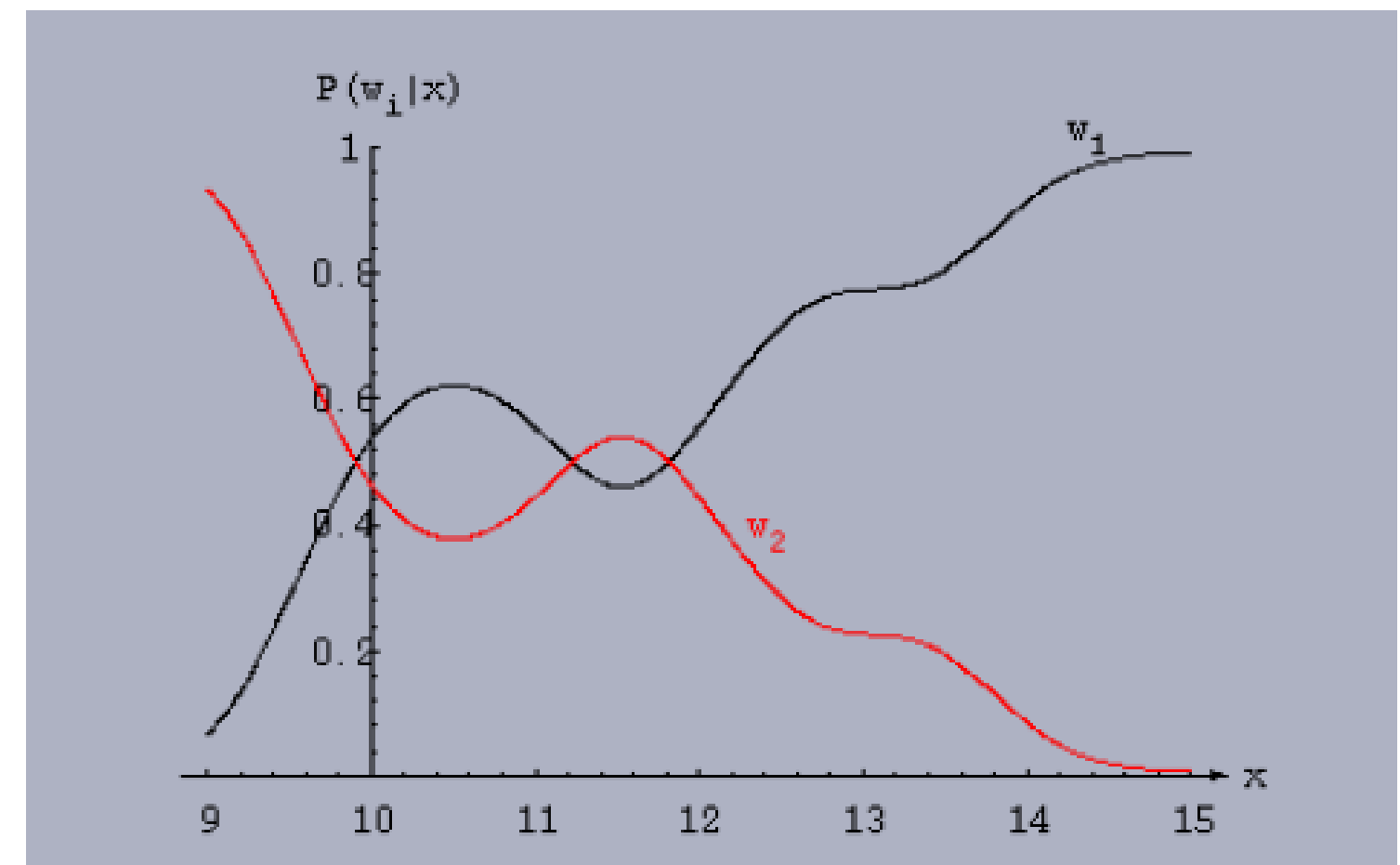
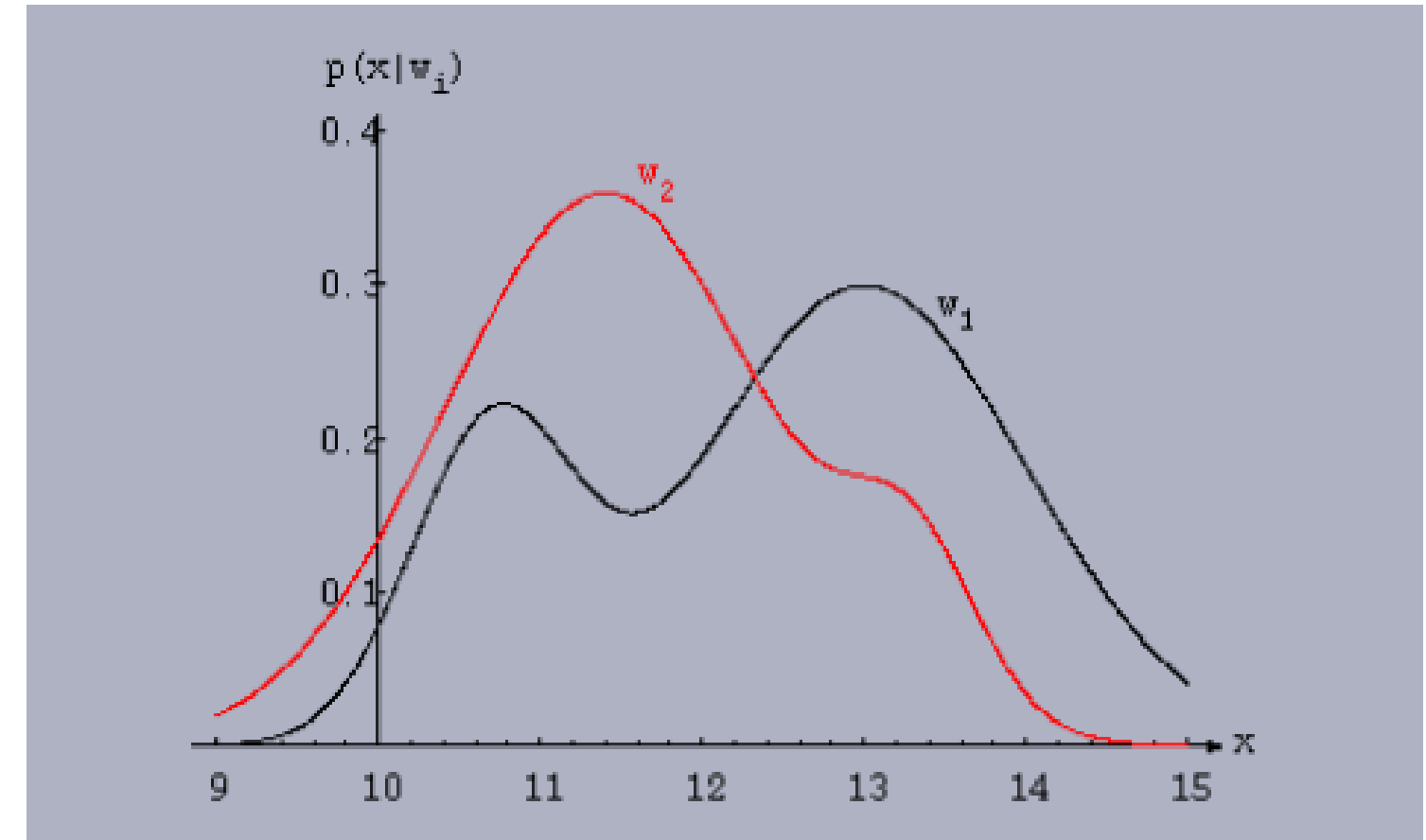
Inference(Reasoning) for a Question

If $x = x_1$, then $y = y_1$

If $x = x_2$, then $y = y_2$

$$p(y = y_j | x = x_i) = \frac{(p(x = x_i | y = y_j)p(y = y_j))}{p(x)},$$
$$p(x) = \sum_i p(x = x_i | y = y_j)p(y = y_j)$$

Density Estimation



Bayesian Theory

Artificial Intelligence

Deep Neural Networks for Learning and Inference

$$o = f(W, x), \text{ ex, } o_j = p(y = y_j | x = x_i)$$

Training (learning)

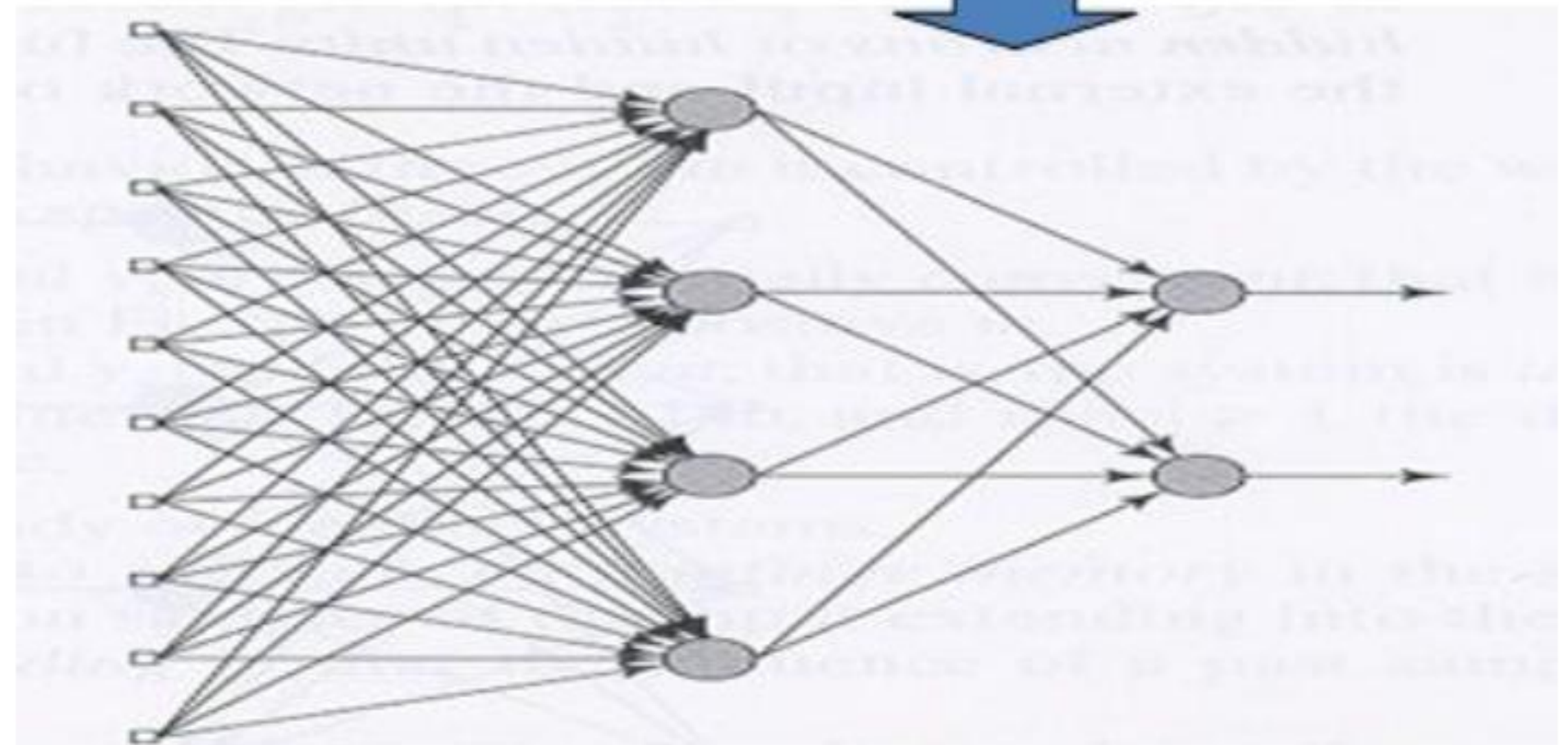
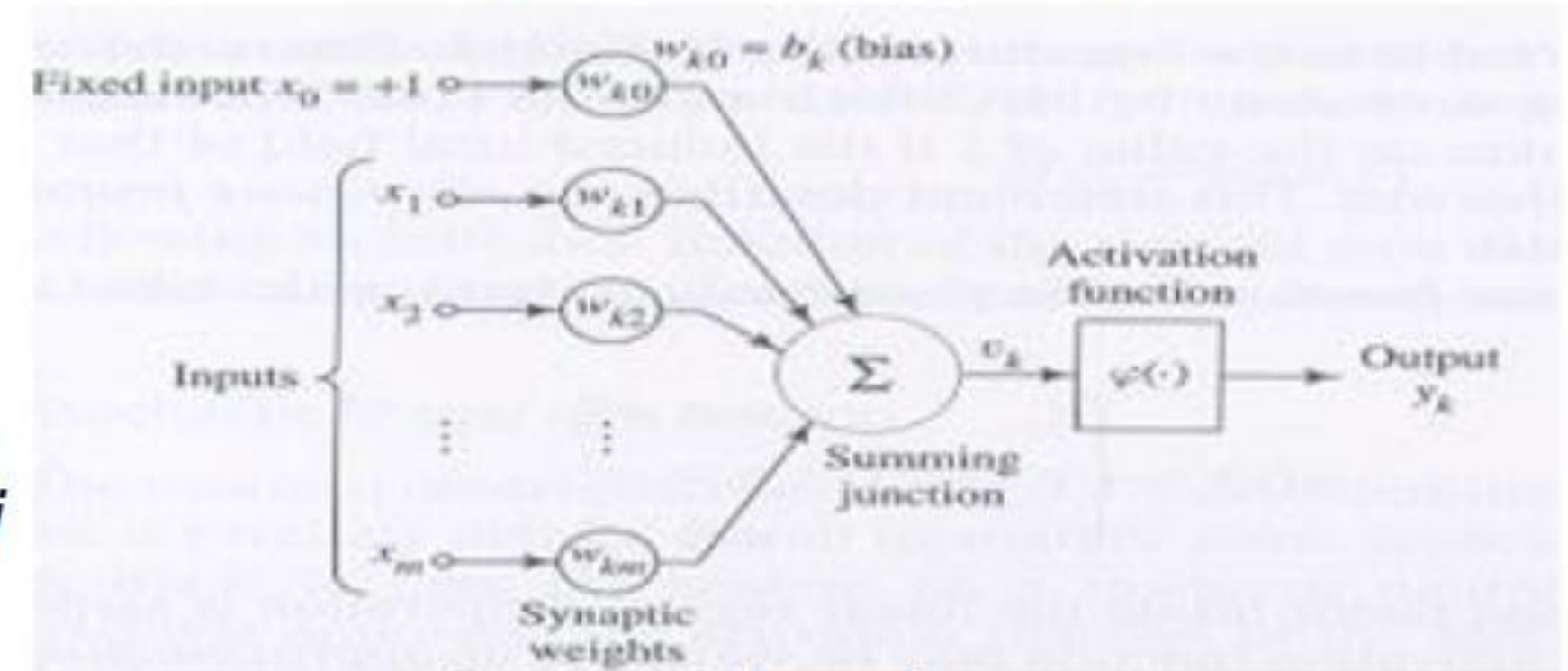
Find W to minimize the errors between o_j and d_j
for given training data $\{(x_p, l_p)\}$

Inference(Reasoning)

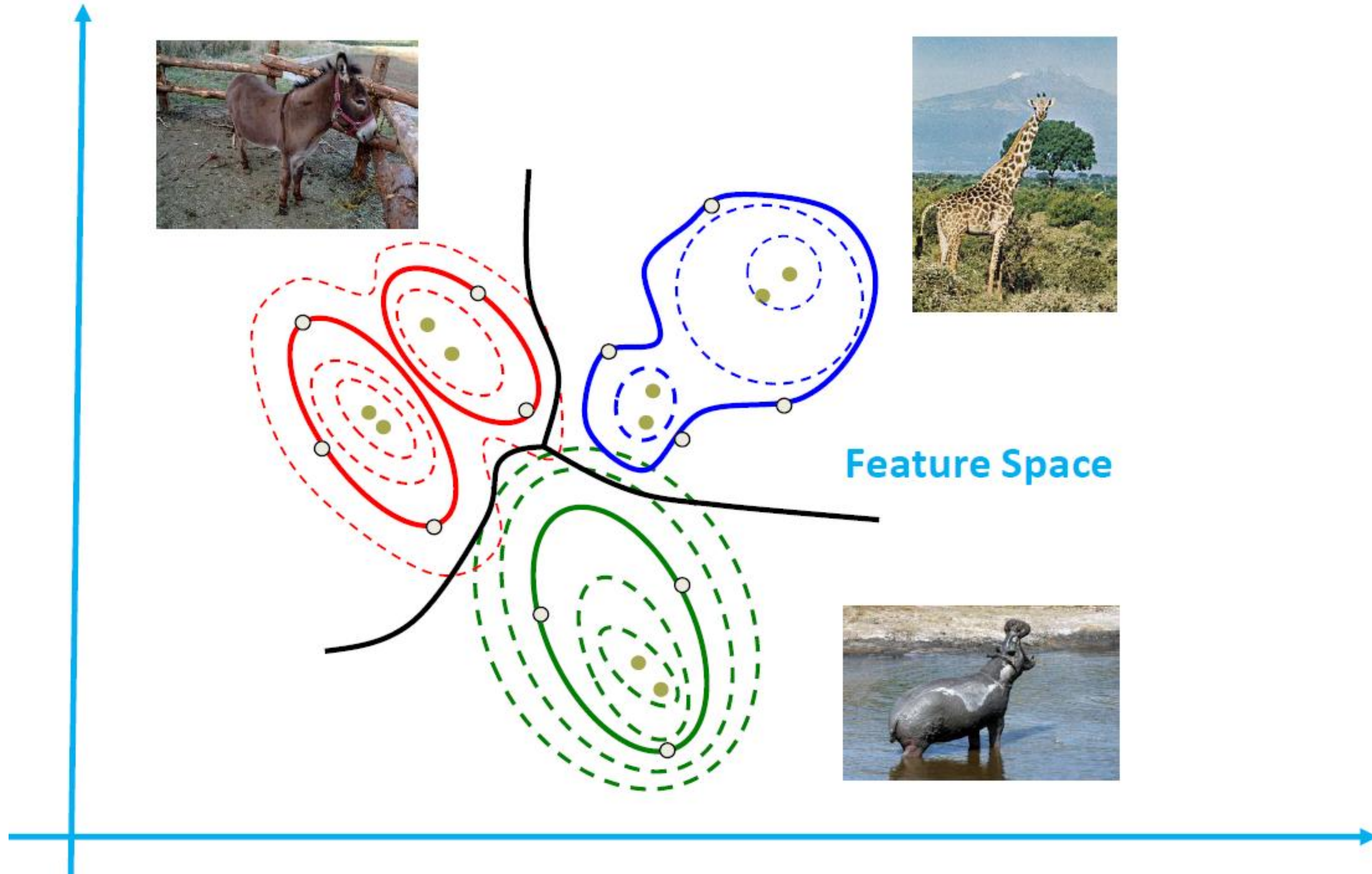
Calculate o_j via the deep network

Connectionism

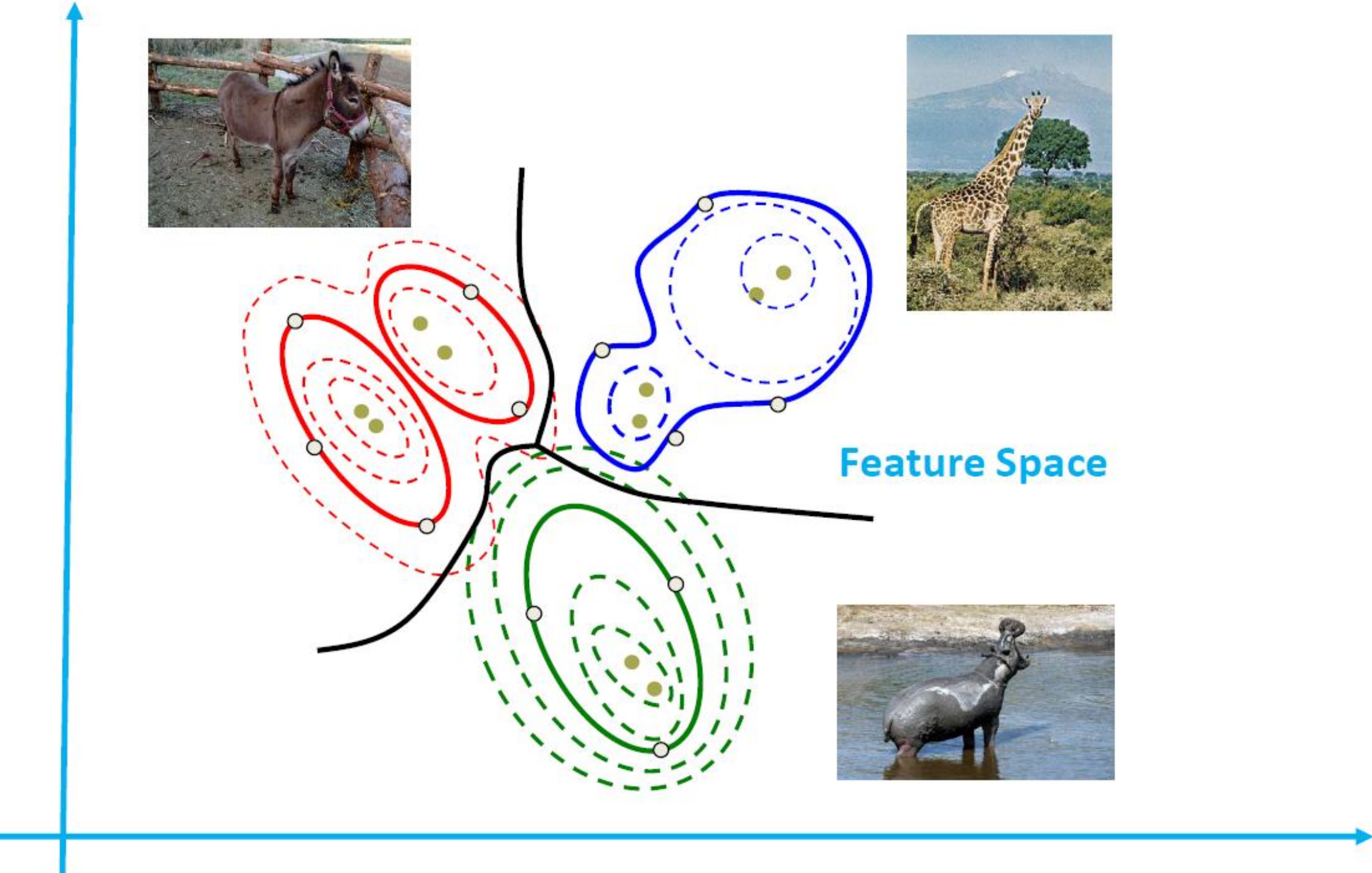
Network Training Inference (feedforward)



Learning and Inference

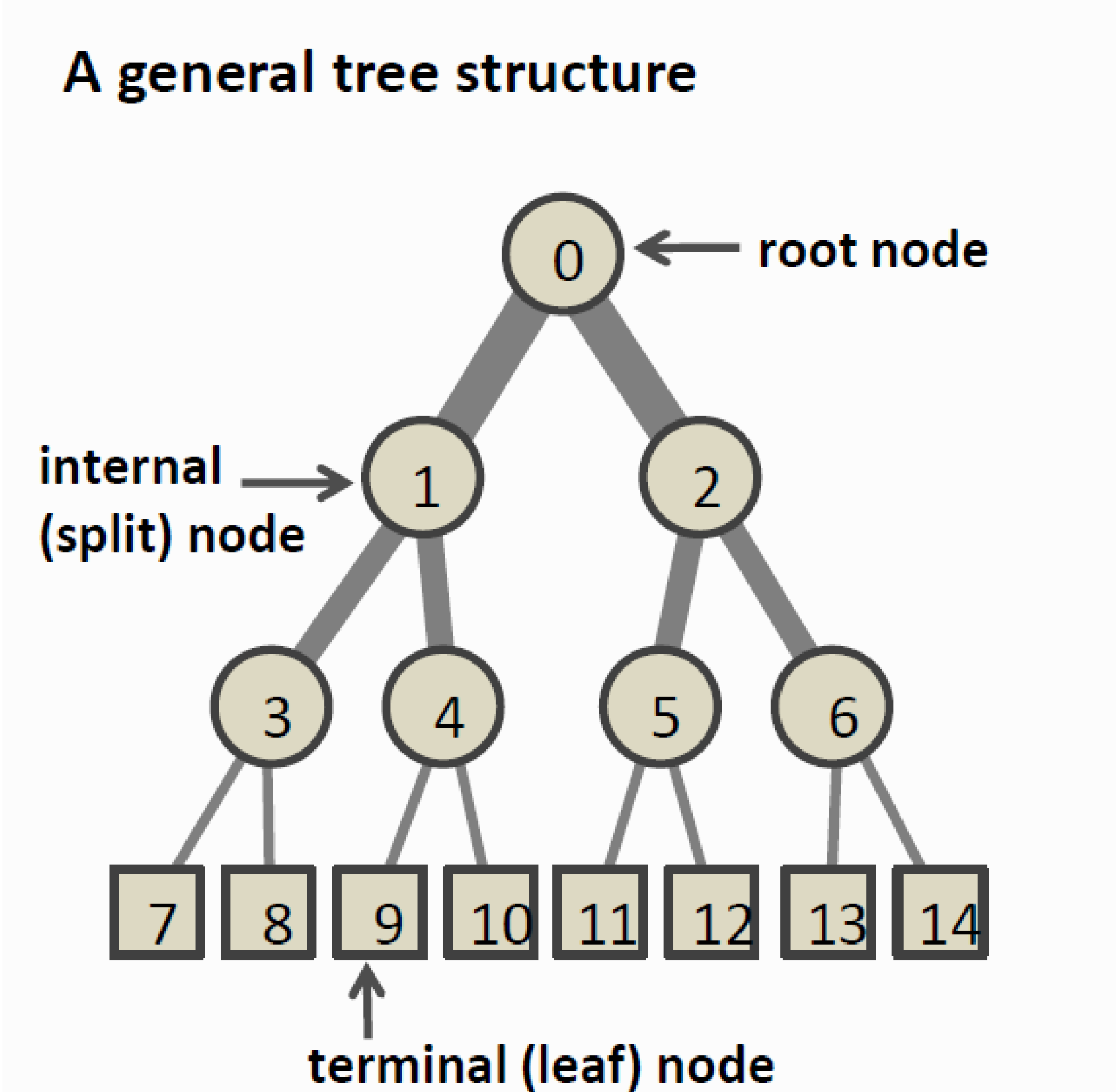


Learning and Inference



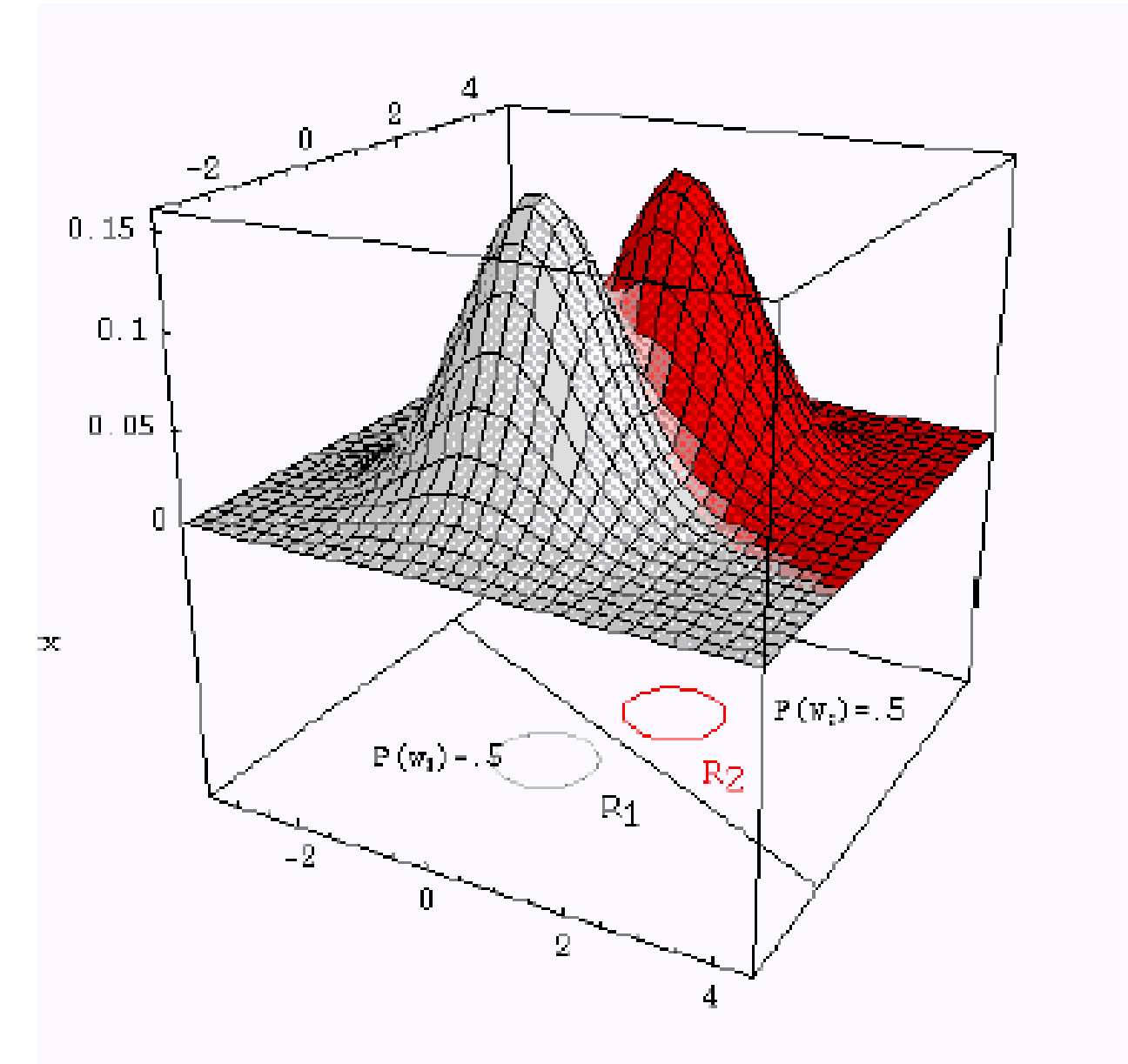
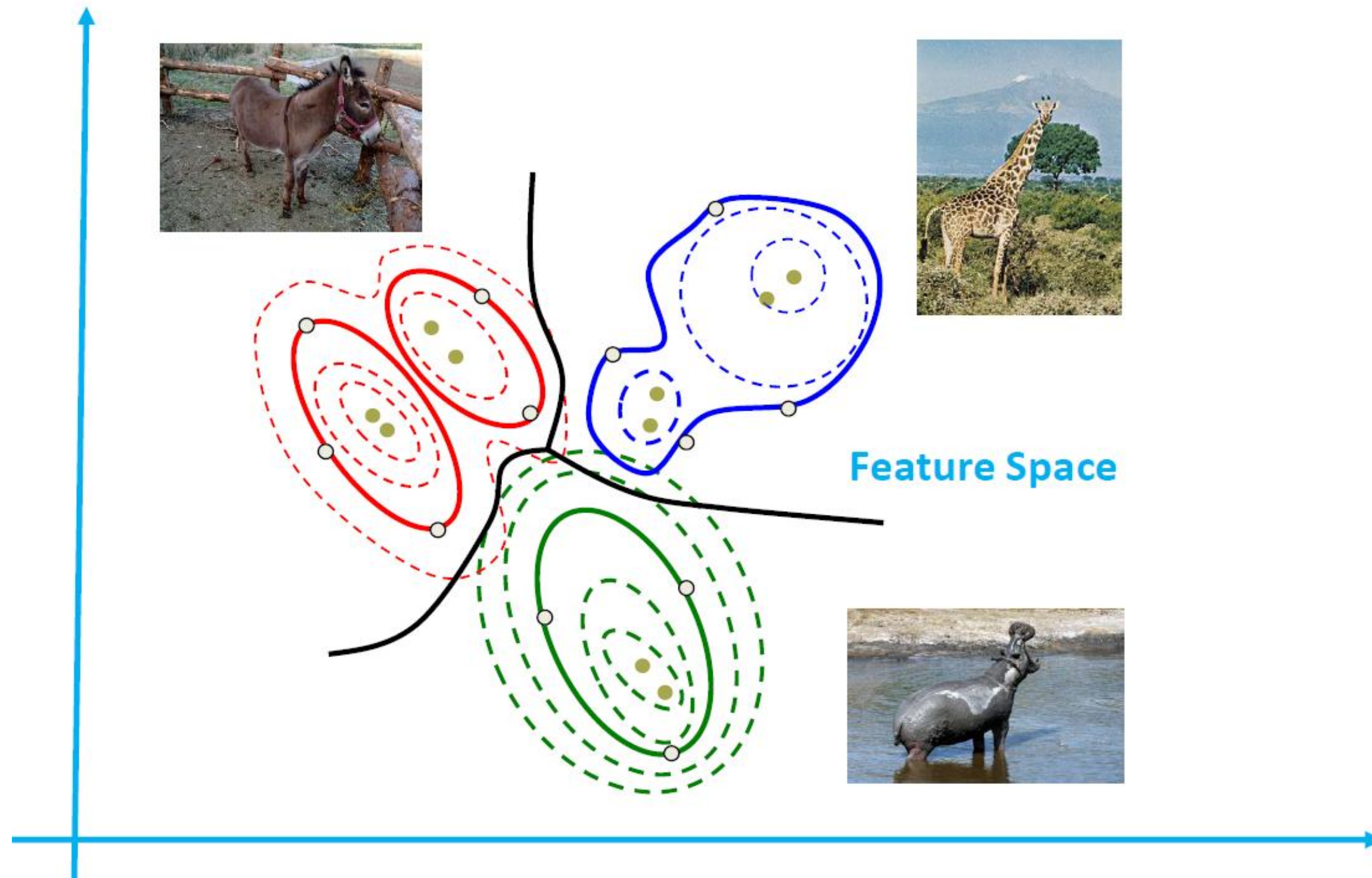
y	x					
y_1	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	x_{16}
y_2	x_{21}	x_{22}	x_{23}	x_{24}	x_{25}	x_{26}

Symbolism



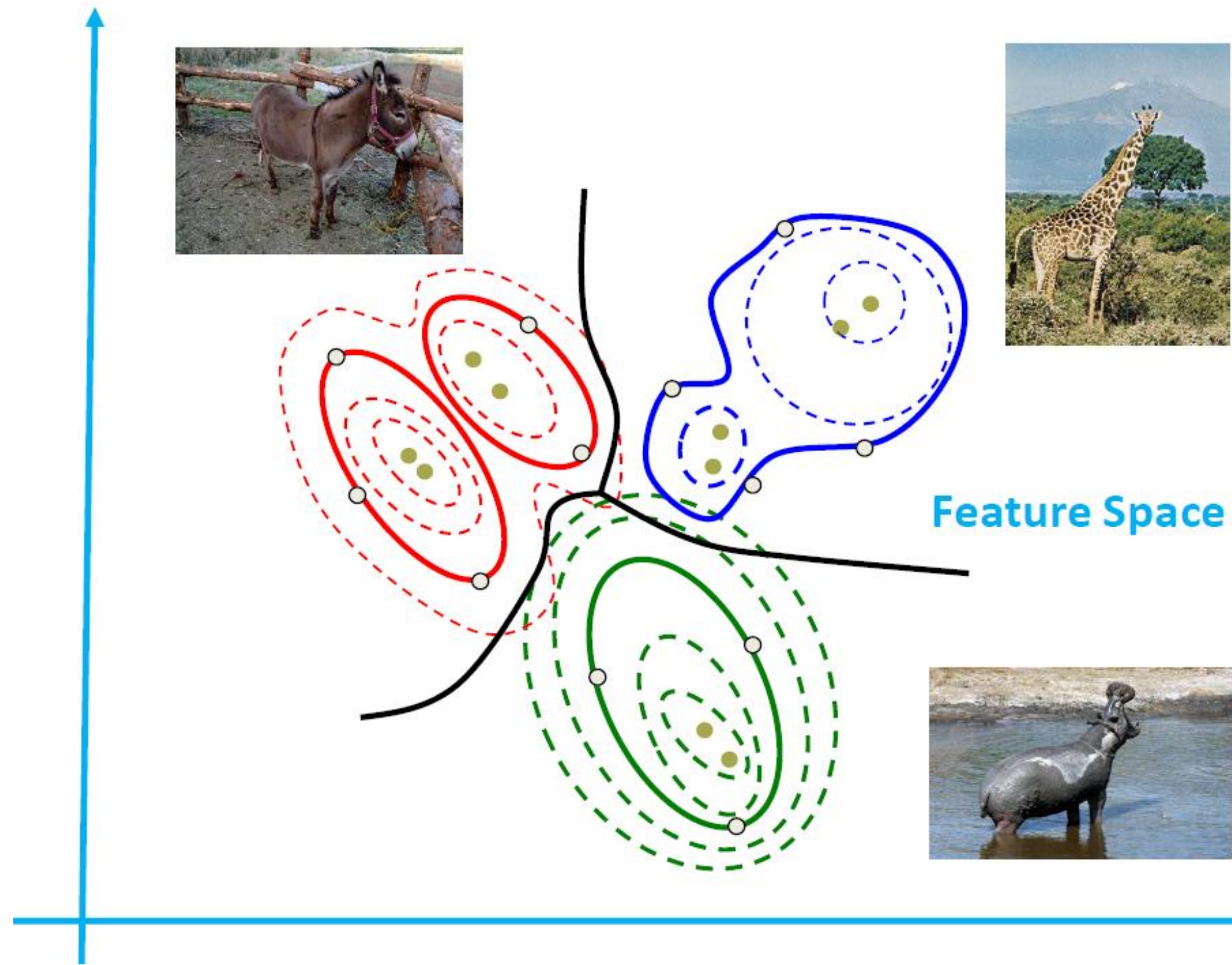
DB
Decision Tree

Learning and Inference



$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]$$

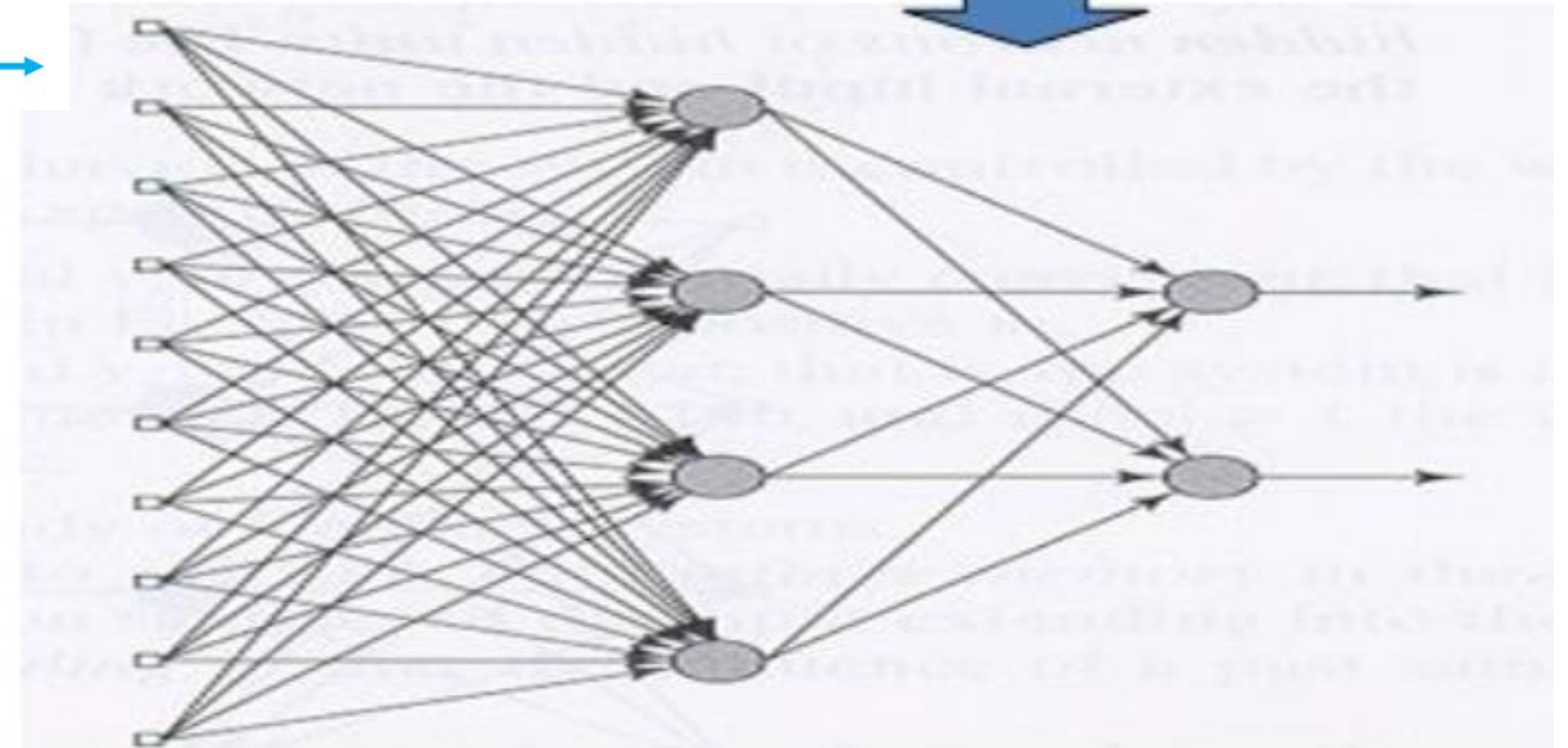
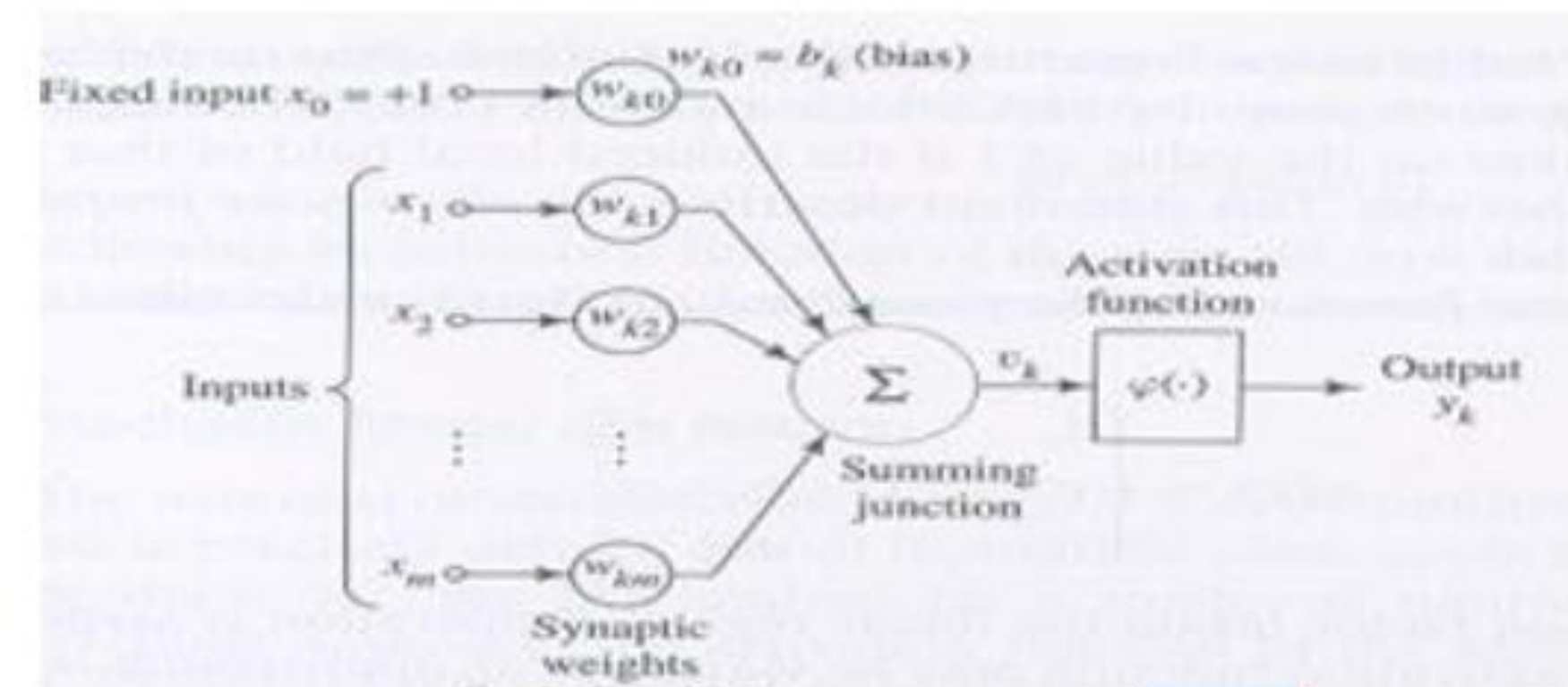
Learning and Inference



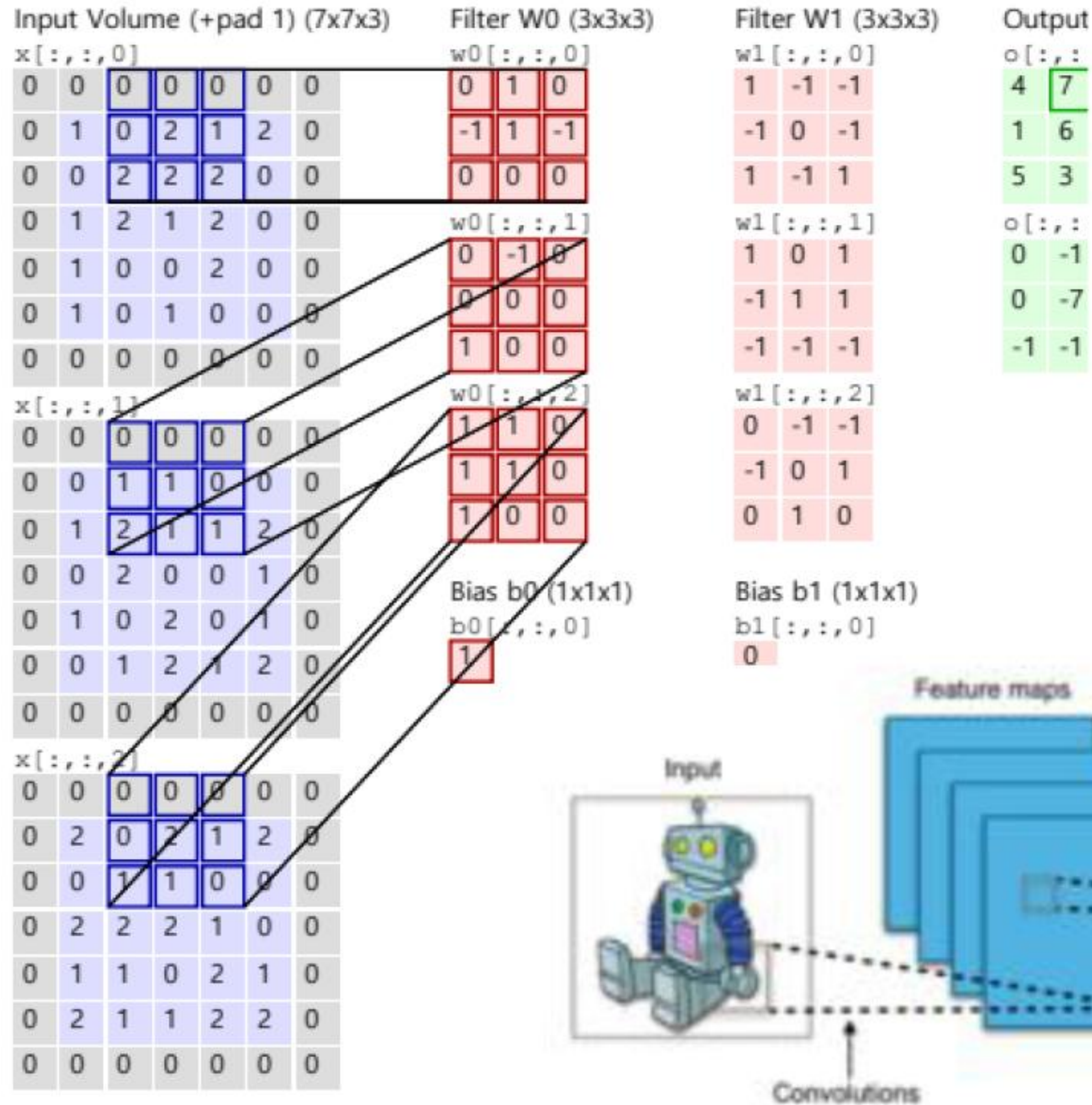
$$g_i(\mathbf{x}) = \mathbf{w}_i^t \mathbf{x} + w_{i0},$$

Connectionism

Network Training
Inference (feedforward)

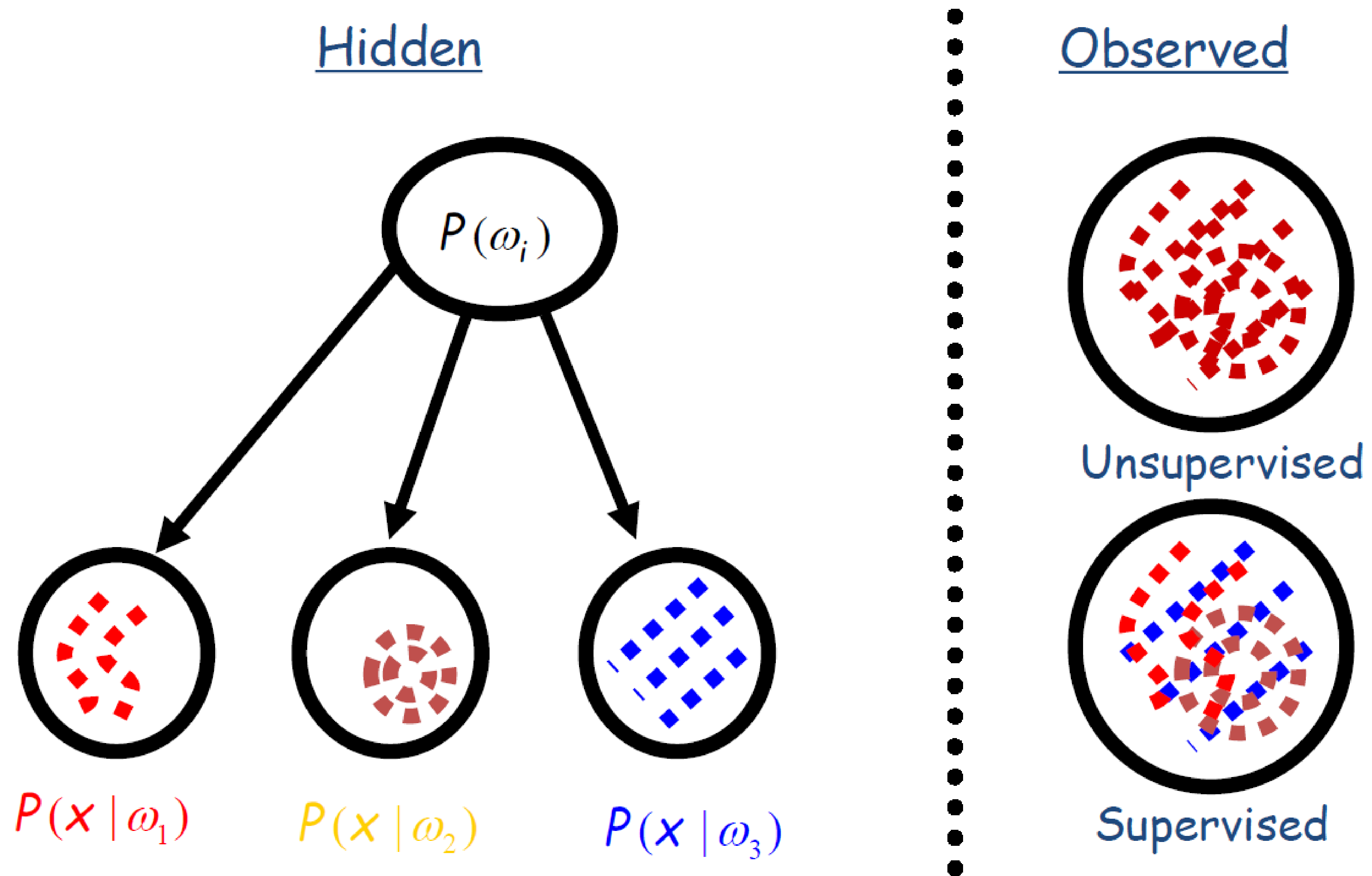


Convolutional Neural Networks



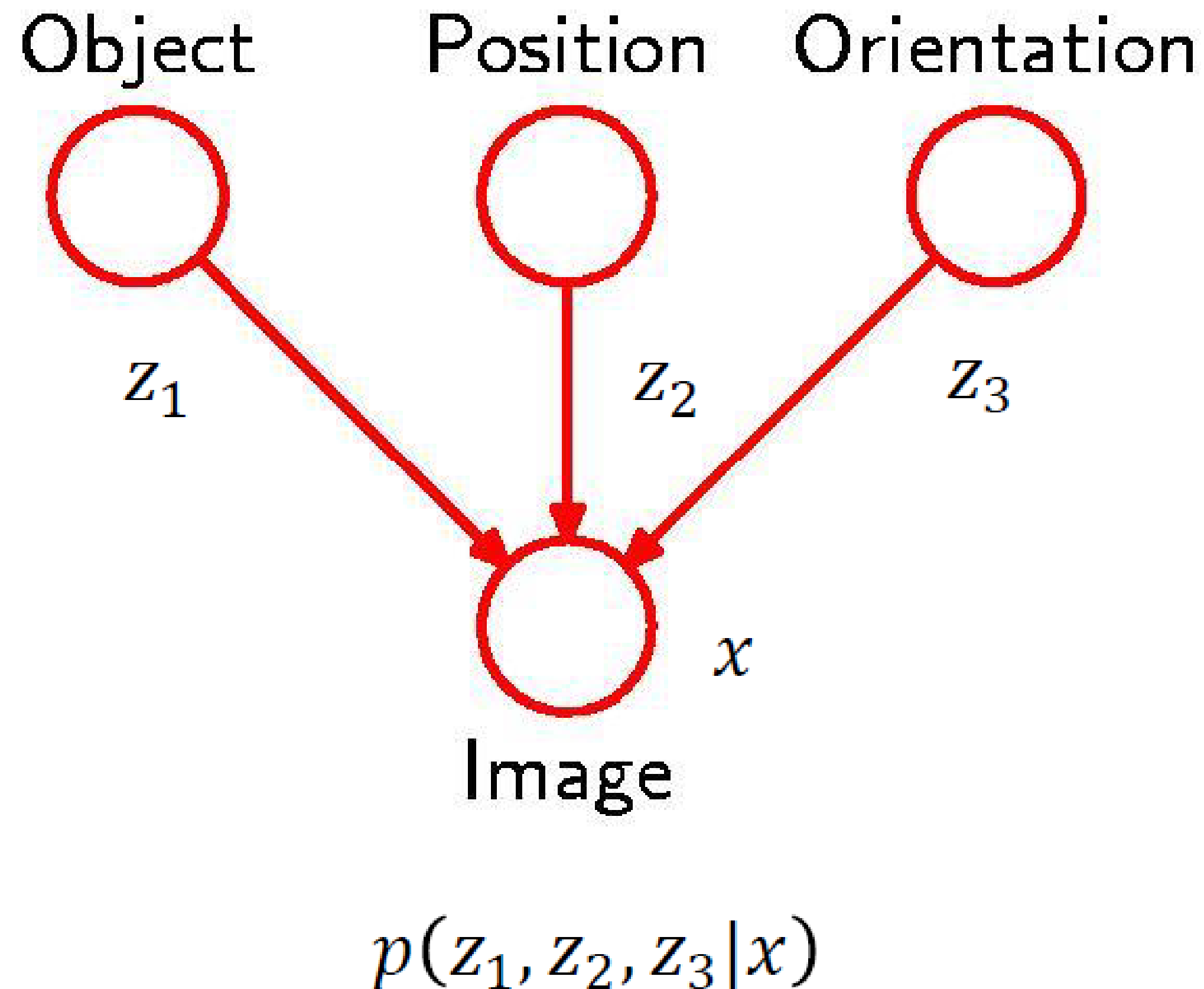
$$o = f(W, x)$$

Supervised/Unsupervised Learning



Generative/Discriminative Model

- Generating images



Generative approach:

Model $p(z, x) = p(x|z)p(z)$

Use Bayes' theorem

$$p(z|x) = \frac{p(x|z)p(z)}{p(x)}$$

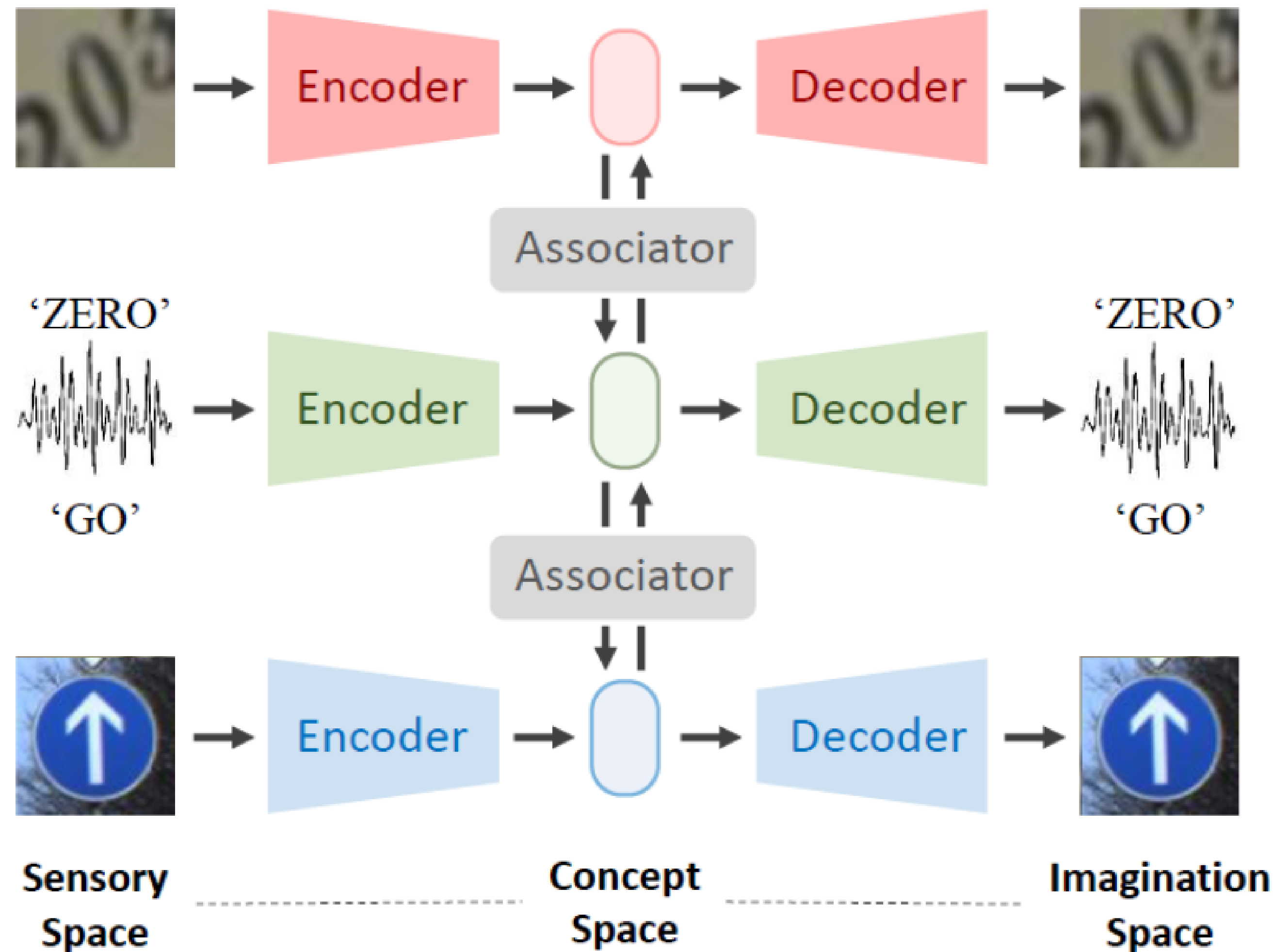
Discriminative approach:

Model $p(z|x)$ directly

Unsupervised Learning

Clustering: K-means, etc.

Variational Auto-Encoder



Statistical Learning

L_2 Loss

$$L(d, f(W, x)) = \|d - f(W, x)\|_2^2$$

Total Loss

$$\mathcal{L}(W) = \int L(d, f(W, x)) dp(x, d)$$

where $p(x, d)$ is a joint PDF of x and d , but unknown

Empirical Total Loss

$$\mathcal{L}_i(W) = \frac{1}{N} \sum_{n=1}^N L(d_n, f(W, x_n))$$

Statistical Learning

$$I(x_k) = \log\left(\frac{1}{p_k}\right) \rightarrow \begin{cases} \text{base 2} \rightarrow \text{bits} \\ \text{base } e \rightarrow \text{nats} \end{cases}$$

32 bits $\rightarrow p_k = 1/2^{32}$ for uniform distribution $\rightarrow I_k = 32$ bits

- ① $I(x_k) = 0$ for $p_k = 1$
- ② $I(x_k) \geq 0$ for $0 \leq p_k \leq 1$
- ③ $I(x_k) \geq I(x_j)$ for $p_k \leq p_j$

Entropy : a measure of the *average amount of information conveyed per message*, i.e., expectation of Information

$$H(x) = E[I(x_k)] = \sum_k p_k I(x_k) = - \sum_k p_k \log p_k$$

Statistical Learning

Entropy becomes maximum when p_k is equiprobable

32 bits $\rightarrow p_k = 1/2^{32}$ for uniform distribution $\rightarrow I_k = 32$ bits

$$\rightarrow 0 \leq H(X) \leq -\sum_{k=1}^{2^{32}} \frac{1}{2^{32}} \log \frac{1}{2^{32}} = 32$$

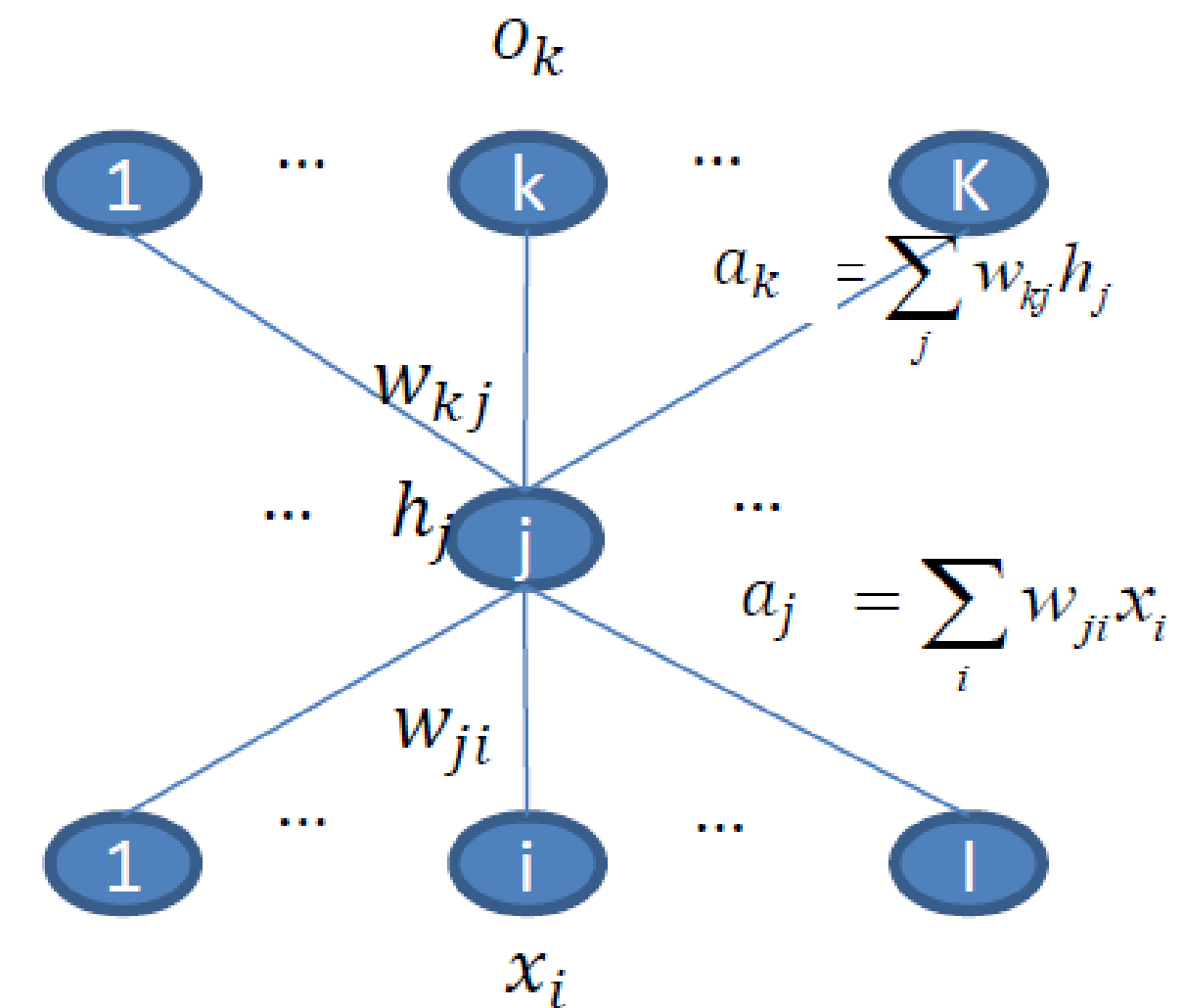
$\rightarrow H(X) = 0$ for an event $p_k = 1$ and $p_{j \neq k} = 0$

Cross Entropy Loss:

$$\mathcal{L}(W) = -\sum_k^K t_k \log f_k(W, x)$$

$$f_k(W, x) = \frac{e^{a_k}}{\sum_j e^{a_j}} \text{ (softmax)}$$

t_k : target label (one hot: 0000100)



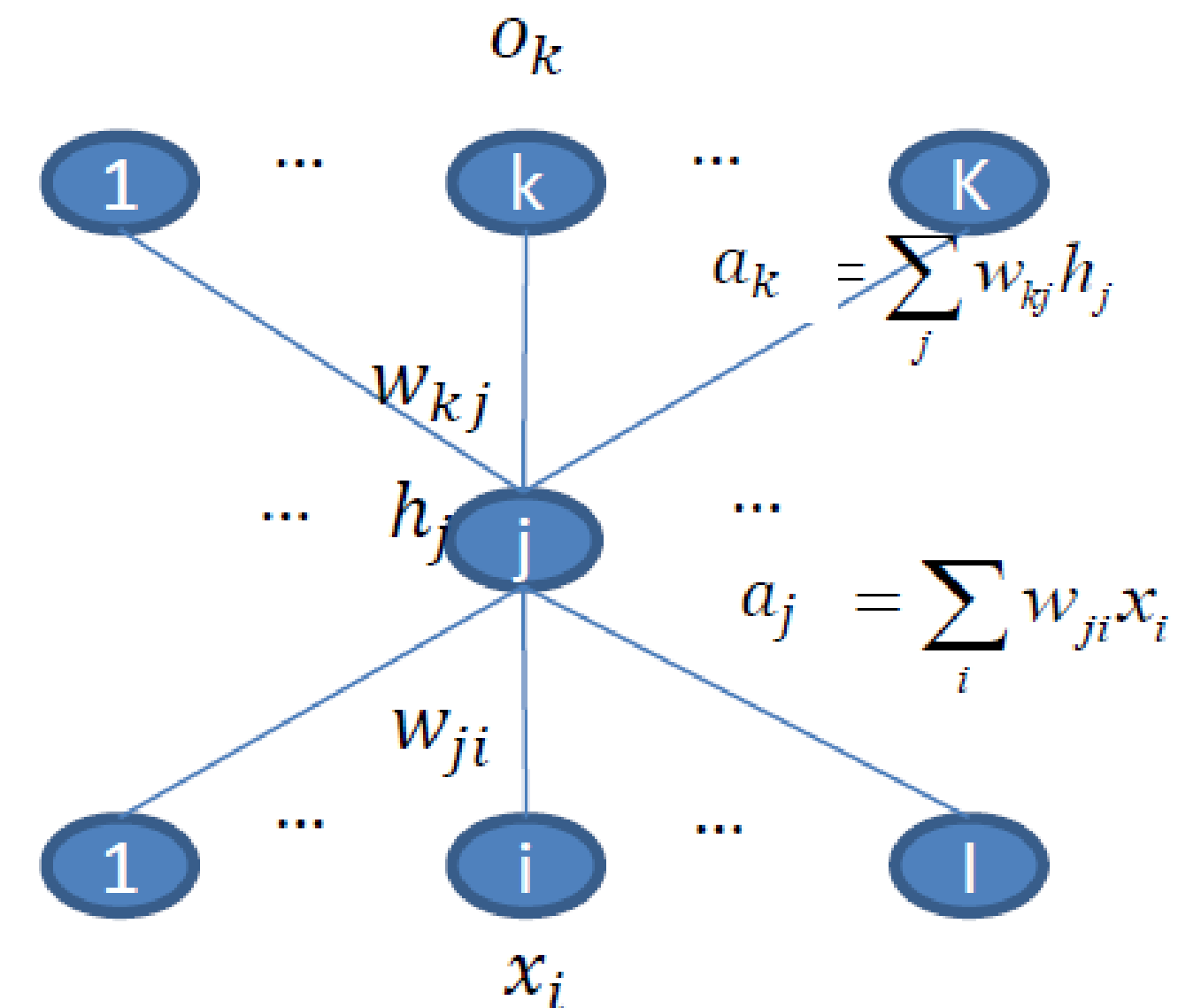
Statistical Learning

Cross Entropy Loss:

$$\mathcal{L}(W) = - \sum_k^K [t_k \log f_k(W, x) + (1 - t_k) \log (1 - f_k(W, x))]$$

$$f_k(W, x) = \frac{1}{1 + e^{-a_k}} (\text{sigmoid})$$

t_k : target label (multi-hot: 00110100)



Statistical Learning

Theorem (Gray 1990)

$$\sum_k p_k \log \frac{p_k}{q_k} \geq 0$$

Relative entropy (or **Kullback – Leibler divergence**)

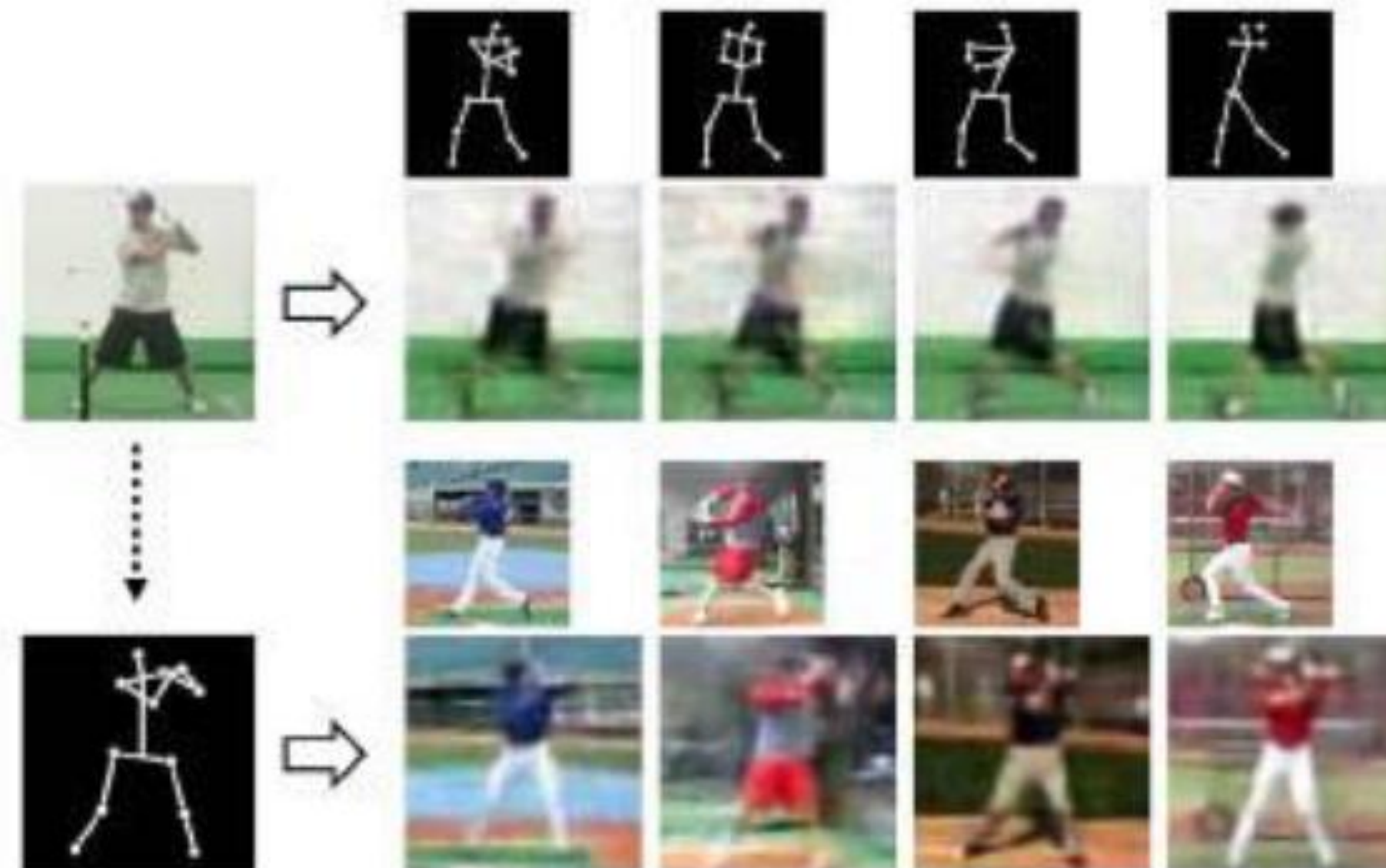
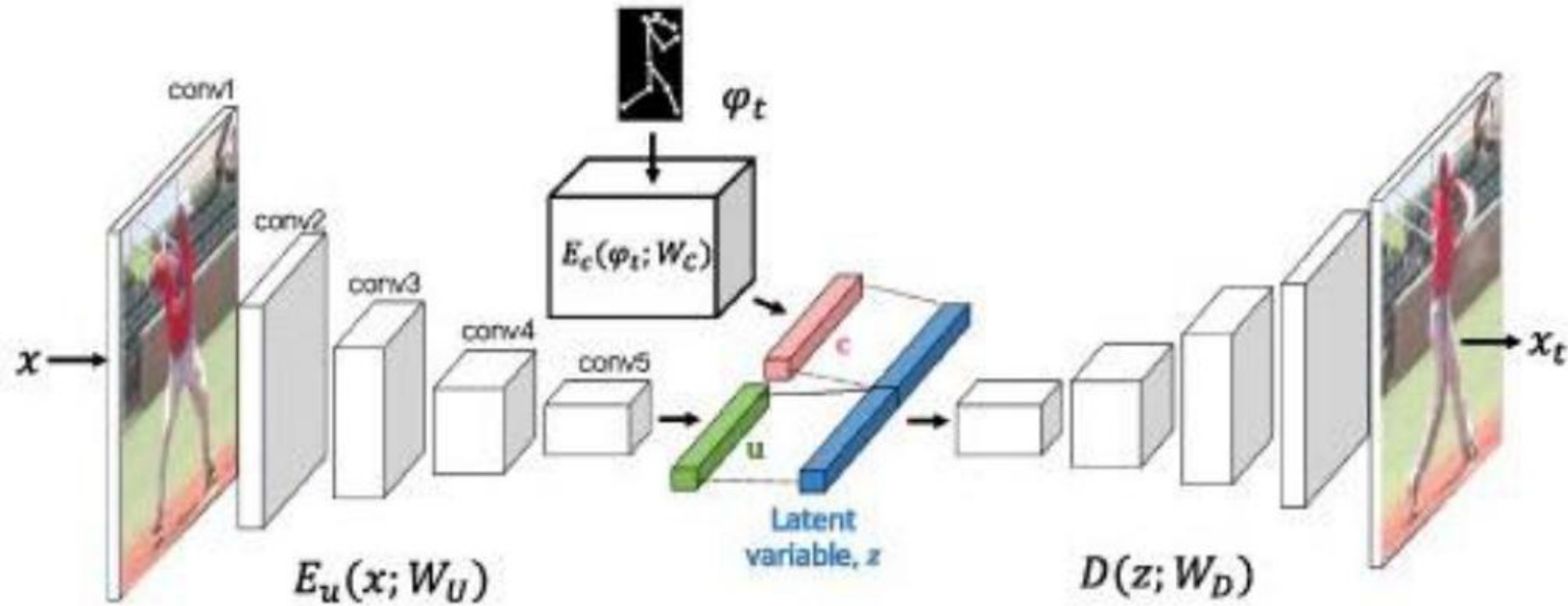
$$D_{KL}(p \parallel q) = \sum_k p_k \log \frac{p_k}{q_k}$$

$$D_{KL}(p \parallel q) = 0 \text{ for } p \equiv q$$

p_k probability mass function

q_k reference probability mass function

Scene and Object Generation



Pose Transformer

Scene and Object Generation

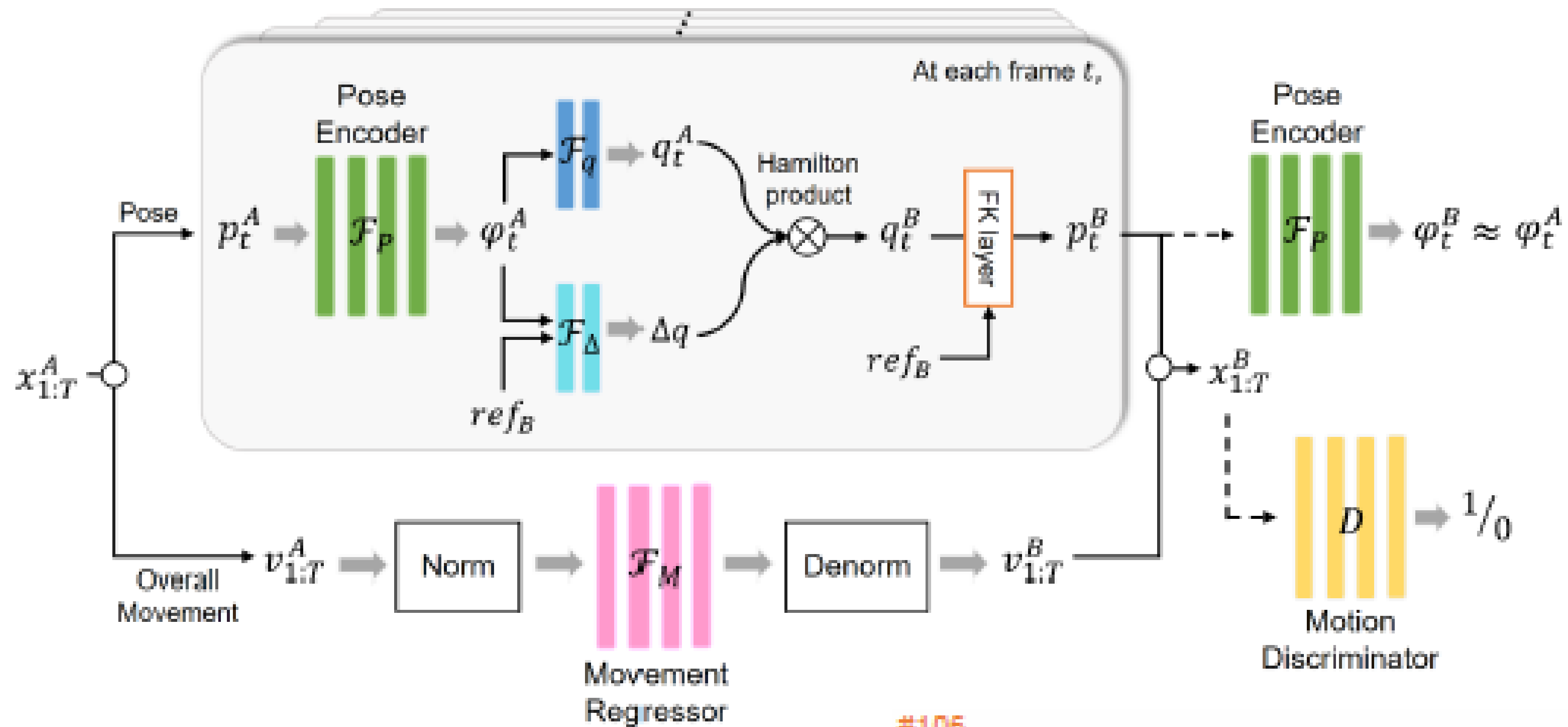
$$\mathcal{L}(\theta, \phi) = \mathcal{L}_{ref} + \mathcal{L}_{pose} + \mathcal{L}_{id}.$$

$$\begin{aligned}\mathcal{L}_{ref} = & \mathbb{E}_{q_{\phi}(z|x_a^k, \varphi_a^k)} [\log p_{\theta}(x_a^k|z)] \\ & - D_{KL} \left(q_{\phi}(z|x_a^k, \varphi_a^k) \parallel p_{\theta}(z) \right) .\end{aligned}$$

$$\begin{aligned}\mathcal{L}_{pose} = & \mathbb{E}_{q_{\phi}(z|x_a^k, \varphi_a^l)} [\log p_{\theta}(x_a^l|z)] \\ & - D_{KL} \left(q_{\phi}(z|x_a^k, \varphi_a^l) \parallel p_{\theta}(z) \right) \\ & - \lambda_u \cdot D_{KL} \left(q_{\phi}(u|x_a^l) \parallel q_{\phi}(u|x_a^k) \right) .\end{aligned}$$

$$\begin{aligned}\mathcal{L}_{id} = & \mathbb{E}_{q_{\phi}(z|x_b^{k'}, \varphi_a^k)} [\log p_{\theta}(x_b^{k'}|z)] \\ & - D_{KL} \left(q_{\phi}(z|x_b^{k'}, \varphi_a^k) \parallel p_{\theta}(z) \right) \\ & - \lambda_c \cdot D_{KL} \left(q_{\phi}(c|\varphi_b^{k'}) \parallel q_{\phi}(c|\varphi_a^k) \right)\end{aligned}$$

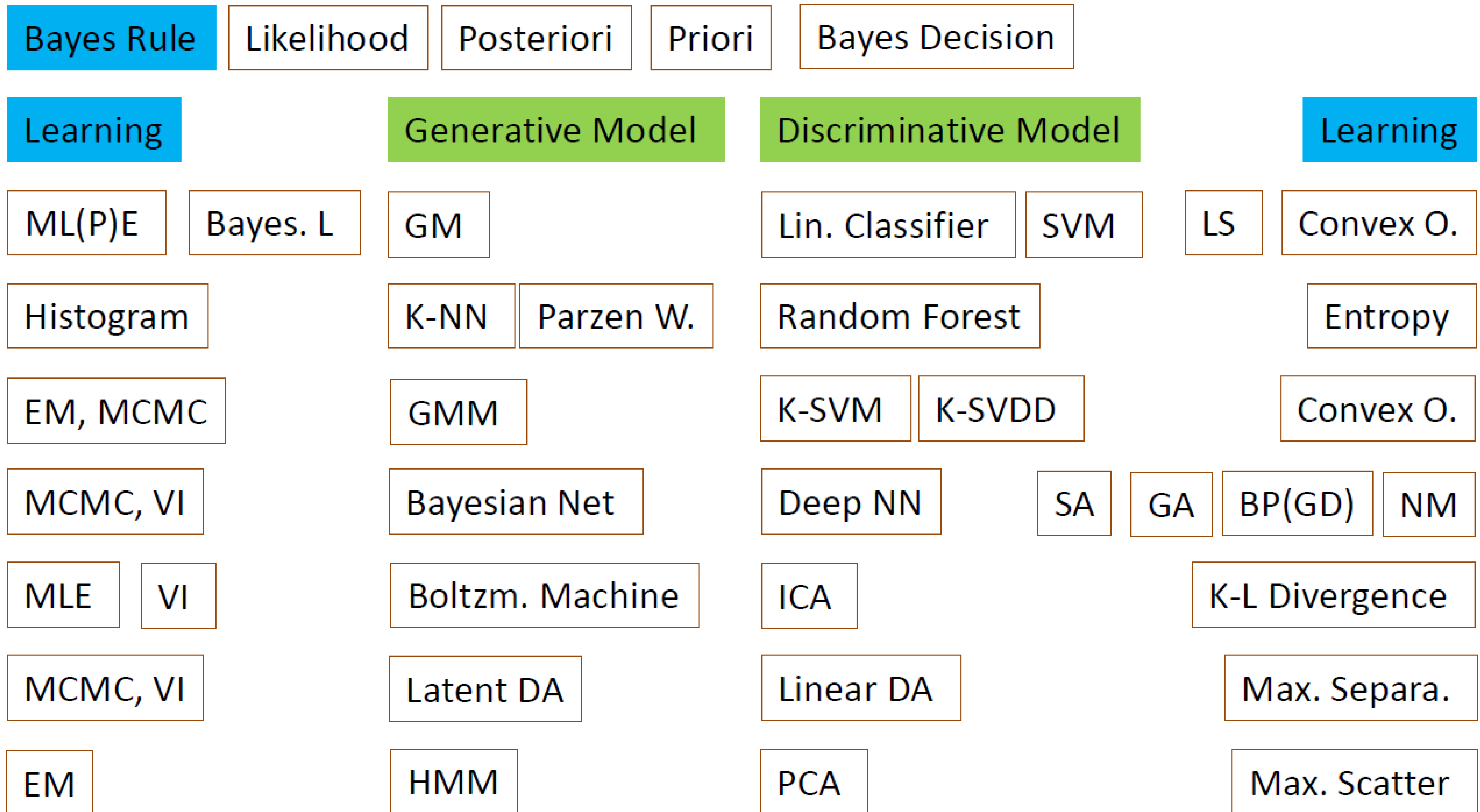
Motion Retargeting



#106



Outline of ML Techniques



Course Outline

Intro. AI, ML, and DL

Intro. Linear Algebra

Intro. Prob. & Information

Bayesian Decision Theory

Dim reduction PCA & LDA

Learning Rules

Support Vector Machine

Deep Convolutional Networks

Bayesian Networks

Parametric pdf Estimation

Non-Parametric pdf Estimation

Boltzman Machine

Markov Chain Monte Carlo

Inference of Bayesian Net, MCMC

Inference of Bayesian Net, VI

Traffic Pattern Analysis, VI

Recent Papers

- Active Learning
- Imbalanced Data Learning
- Out of Distribution
- Weakly Supervised Learning
- Etc.